SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Data Warehousing and Data Mining
COURSE CODE:CSA1609

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items ***(2)***	Comparison Depth(2.5)	Use of Evidence ***(2)***	Critical Thinking ***(2)***	Clarity of Presentation ***(1)***	***Total(10)***
Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I G Madhulatha(192311295)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
G Madhulatha

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation

Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: **Assessment Tool 3: Comparative Analysis (25% → 5 Marks)**

Time duration :45 Minutes

Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer	Income (₹)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. (BL4 – 1 Mark)

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)

Q4. Dataset: Retail Products

Product	Sales (₹)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)

Q5 . Dataset: Mixed Attributes

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7

I4	250	B	3.5
I5	520	A	4.6

Question:
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)

ANSWERS

Q1. Customer Data

Question
Compare K-Means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)
Dataset

Customer	Income (₹)	Spending Score
C1	30,000	60
C2	60,000	40
C3	50,000	70
C4	28,000	65
C5	65,000	35

Comparative Analysis

Feature	K-Means	Hierarchical Clustering
Approach	Partitions data into K clusters	Builds a hierarchy of clusters
Cluster selection	Requires K in advance	Dendrogram helps select clusters
Speed	Generally faster	Generally slower
Scalability	Good for large datasets	Less scalable for very large datasets
Result	Gives final clusters	Gives a hierarchy of clusters

For this dataset, **K-Means with K = 2** can identify two broad customer groups:

- **C1, C4** → lower-income but higher-spending customers
- **C2, C3, C5** → higher-income customers with different spending behavior

Hierarchical clustering can provide more information about how individual customers progressively merge into groups.

Conclusion
For this small dataset, **both methods are suitable**, but **K-Means is faster and more scalable**, while hierarchical clustering is more useful when the organization wants to study the hierarchy and relationships among customers.

Q2. Document Data

Question
Compare K-Means and EM algorithms in handling overlapping clusters for the given dataset. (BL4 – 1 Mark)
Dataset

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Comparative Analysis

Feature	K-Means	EM
Type	Hard clustering	Soft/probabilistic clustering
Assignment	One document belongs to one cluster	Document can have probabilities for multiple clusters
Overlapping clusters	Less effective	More effective
Output	Cluster label	Probability of membership
Model	Distance-based	Probability/distribution-based

The dataset shows two obvious groups:

- **D1, D2, D5** have relatively high Word2/Word3 values.

- **D3, D4** have very high Word1 values.

K-Means could assign each document to one cluster.
For example:
D1 → Cluster 1
However, if a document has characteristics of both groups, K-Means must still assign it to only one cluster.
EM can represent uncertainty. For example, an illustrative result could be:
D1 → 80% Cluster 1, 20% Cluster 2
The exact probabilities require running EM and are not determined directly from the given table.

Conclusion
For **overlapping document clusters, EM is more suitable than K-Means** because EM provides probabilistic membership. K-Means is simpler and faster but uses hard assignments.

Q3. Geographic Data

Question
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)
Dataset

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

1. Euclidean Distance
Euclidean distance is the straight-line distance between two points.

For example, between P1 and P2:

2. Manhattan Distance
Manhattan distance measures the total absolute difference along each dimension.

For P1 and P2:

Comparative Analysis

Feature	Euclidean	Manhattan
Calculation	Straight-line distance	Sum of absolute differences
Formula		(\sum
Effect of large differences	More sensitive	Less sensitive
Path interpretation	Direct distance	Grid/city-block distance
Common use	Geographical/spatial clustering	High-dimensional or grid-like data

For this dataset, nearby points such as **P1 and P2** are likely to remain close under both metrics. However, the numerical distances and potentially some borderline cluster assignments can change depending on the metric.
Important practical point
Latitude and longitude are geographic coordinates. For actual geographical distance over larger areas, using raw Euclidean distance on latitude/longitude is only an approximation. A geographic distance such as **Haversine distance** is generally more appropriate when accurate surface distance is required.

Conclusion
Euclidean distance is suitable when straight-line similarity is required, while **Manhattan distance** is useful when differences along individual dimensions should be added directly. The choice of metric can change distance values and therefore affect cluster membership, especially for points near cluster boundaries.

Q4. Retail Products

Question
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)
Dataset

Product	Sales (₹)	Frequency
P1	50,000	20
P2	30,000	10
P3	52,000	22
P4	25,000	8
P5	32,000	12

1. Agglomerative Clustering
Agglomerative clustering is a **bottom-up** approach.
Initially:
P1 | P2 | P3 | P4 | P5
Each product is treated as an individual cluster.
The algorithm progressively merges similar products.
For this dataset:

- P1 and P3 are highly similar.
- P2 and P5 are relatively similar.
- P4 is closer to the lower-sales/lower-frequency group.

A possible grouping is:
Cluster 1: P1, P3
Cluster 2: P2, P4, P5

2. Divisive Clustering
Divisive clustering is a **top-down** approach.
Initially:
All products → One cluster
The algorithm repeatedly splits the cluster into smaller groups.
For example:
All products
↓
High-sales products / Lower-sales products
↓
Further subdivisions
Comparative Analysis

Feature	Agglomerative	Divisive
Approach	Bottom-up	Top-down
Starting point	Individual products	One large cluster
Operation	Merge clusters	Split clusters
Complexity	Generally practical and commonly used	Often more computationally demanding
Dataset suitability	Very suitable for small datasets	Can be used, but unnecessary here

Application to the Dataset
The dataset contains only **5 products**, so both approaches can be used.
However, **agglomerative clustering is more suitable** because it easily reveals which products are similar.
For example:

- P1 and P3 → high sales and high frequency
- P2, P4 and P5 → lower sales and frequency

Conclusion
For this small retail dataset, **agglomerative clustering is more suitable** because it is simple to apply and clearly shows how similar products merge into groups. Divisive clustering is also possible but is generally less convenient for this small dataset.

Q5. Mixed Attributes

Question
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)
Dataset

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

The dataset contains:

- **Price** → numerical
- **Category** → categorical

- **Rating** → numerical

Therefore, it is a **mixed-type dataset**.

1. Centroid-Based Clustering
K-Means is a common centroid-based algorithm.
It calculates a centroid for each cluster and assigns objects according to distance from the centroid.
Problem with this dataset
Standard K-Means works naturally with numerical data.
The **Category** attribute cannot simply be treated as an ordinary number because:
Category A = 1 and Category B = 2
does not necessarily mean that B is "greater" than A.
Therefore, categorical information needs appropriate encoding or a clustering method designed for mixed data.
Based on the numerical characteristics:

- I1, I3, I5 have high prices and high ratings.
- I2, I4 have lower prices and lower ratings.

The natural grouping is:
Cluster 1: I1, I3, I5
Cluster 2: I2, I4

2. Density-Based Clustering
A density-based method such as **DBSCAN** identifies dense regions of data rather than requiring a centroid for each cluster.
It can:

- Identify dense groups.
- Detect outliers.
- Discover irregularly shaped clusters.
- Avoid specifying the number of clusters in advance.

However, **standard DBSCAN also requires a suitable distance representation**, so categorical variables cannot simply be inserted as ordinary numeric values. A mixed-data distance measure or a suitable mixed-data clustering algorithm is required.

Comparative Analysis

Feature	**Centroid-Based**	**Density-Based**
Example	K-Means	DBSCAN
Basic idea	Groups around centroids	Groups dense regions
Requires K?	Yes	No
Outlier detection	Weak	Strong
Cluster shape	Generally suitable for compact clusters	Can identify irregular shapes
Mixed data	Needs suitable encoding/variant	Needs suitable mixed-data distance
Given dataset	Suitable after preprocessing	Possible, but unnecessary for this tiny dataset

Applying to the Given Data
The natural groups are:
Cluster 1
I1, I3, I5
Characteristics:

- Category A
- Price around ₹500–₹550
- Rating around 4.5–4.7

Cluster 2
I2, I4
Characteristics:

- Category B
- Price around ₹250–₹300
- Rating around 3.5–3.8

Which is more effective?
For this particular dataset, **centroid-based clustering is simpler after properly handling the categorical attribute**, because the data naturally forms two compact groups.
However, if the dataset were much larger and contained:

- Irregular clusters

- Noise
- Outliers

then density-based clustering could be more advantageous.
For genuinely mixed numerical and categorical data, methods specifically designed for mixed attributes, such as **k-prototypes**, are often more appropriate than applying ordinary K-Means directly.

Conclusion

Centroid-based clustering is simple and effective for the compact numerical patterns in this dataset, but the categorical **Category** attribute must be handled appropriately. Density-based clustering is better for irregular clusters and outlier detection, but it also requires a suitable distance measure for mixed data. Therefore, for this small dataset, a **mixed-data-aware centroid approach such as k-prototypes** would be a practical choice.

Final Comparative Summary

Question	Comparison	Best Interpretation
Q1	K-Means vs Hierarchical	K-Means is faster and more scalable; hierarchical gives a cluster hierarchy
Q2	K-Means vs EM	EM is better for overlapping/uncertain clusters because it gives probabilities
Q3	Euclidean vs Manhattan	Euclidean measures straight-line distance; Manhattan sums absolute differences
Q4	Agglomerative vs Divisive	Agglomerative is more convenient for this small dataset
Q5	Centroid vs Density	Centroid is simple for compact groups; density-based is stronger for irregular/noisy data; mixed-data methods are preferred for categorical + numerical attributes

One-line answers for quick revision

- **Q1:** K-Means is preferred for scalability, while hierarchical clustering is useful for understanding cluster relationships through a dendrogram.
- **Q2:** EM is better than K-Means for overlapping clusters because it provides probabilistic membership.
- **Q3:** Euclidean measures straight-line distance, whereas Manhattan measures the sum of coordinate-wise differences; the choice can affect borderline cluster assignments.
- **Q4:** Agglomerative clustering is more suitable for this small retail dataset because it progressively combines similar products.
- **Q5:** For mixed numerical and categorical data, ordinary K-Means/DBSCAN need suitable preprocessing or distance measures; **k-prototypes** is a practical mixed-data alternative.